

Community One — DEMO envelope report

Slice: DEMO-SLICE-B Classification: demo-illustrative

Stamp: Research Draft — Not for Construction · DEMO

PINNED DIGESTS (SHA-256)

fixture: b7a5c46d9e58a3dd5758556734e0899eba970296514be7e4ffbe7d079f82e2de

rulebook: c7ae502fd0e600774cd77ce3526e6d0a836f3ebf648879930af64b1f7aa06abb

geometry: 0275a536a4d7da2791e53c979b53b09a186025d73300893380a1ef28c6cb0cb6

DECLARED FIXTURE INPUTS

Site width (E-W): 225 m [fact.site-width] <- site.widthM

Site depth (N-S): 180 m [fact.site-depth] <- site.depthM

Declared north bearing: 0 deg [fact.north-bearing] <- site.northBearingDeg

External access road width (south): 24 m [fact.access-road-width] <- site.accessRoad.widthM

Requested dwelling units (client intent): 500 DU [fact.requested-du] <- site.requestedDwellingUnits

note: Intent only — never a result.

PERMITTED VALUES (DEMO RULE ENTRIES)

Max ground coverage (of gross site area): 30 percent [fact.rule.site-coverage-max] cites DEMO-B-SITE-COVERAGE-MAX

Max density: 25 du-per-acre [fact.rule.density-max] cites DEMO-B-DENSITY-MAX

Max height: 9 metre [fact.rule.height-max] cites DEMO-B-HEIGHT-MAX

Max storeys: 2 storeys [fact.rule.storeys-max] cites DEMO-B-STOREYS-MAX

Periphery setback: 12 metre [fact.rule.setback-periphery] cites DEMO-B-SETBACK-PERIPHERY

Front (access-road) setback: 15 metre [fact.rule.setback-front] cites DEMO-B-SETBACK-FRONT

Primary internal road width: 15 metre [fact.rule.road-width-primary] cites DEMO-B-ROAD-WIDTH-PRIMARY

Secondary internal road width: 12 metre [fact.rule.road-width-secondary] cites DEMO-B-ROAD-WIDTH-SECONDARY

Min open space (of gross site area): 25 percent [fact.rule.open-space-min] cites DEMO-B-OPEN-SPACE-MIN

Parking norm: 1.500 ecs-per-du [fact.rule.parking-ecs-per-du] cites DEMO-B-PARKING-ECS-PER-DU

Min amenity share (of gross site area): 8 percent [fact.rule.amenity-share-min] cites DEMO-B-AMENITY-SHARE-MIN

Min townhouse plot frontage: 8 metre [fact.rule.unit-plot-frontage-min] cites DEMO-B-UNIT-PLOT-FRONTAGE-MIN

Min townhouse plot depth: 18 metre [fact.rule.unit-plot-depth-min] cites DEMO-B-UNIT-PLOT-DEPTH-MIN

Max unbroken row length: 48 metre [fact.rule.row-length-max] cites DEMO-B-ROW-LENGTH-MAX

DERIVED ENVELOPE (EVERY NUMBER CITES ITS FEEDERS)

Gross site area: 40500 sqm [fact.site-area]

from fixture: site.widthM, site.depthM

cites: (none — fixture-only derivation)

note: width x depth

Gross site area: 10.008 acre [fact.site-area-acres]

from fixture: site.widthM, site.depthM

cites: (none — fixture-only derivation)

note: site area / 4046.8564224 sqm-per-acre (physical constant)

Buildable envelope width: 201 m [fact.envelope-width]

from fixture: site.widthM

cites: DEMO-B-SETBACK-PERIPHERY

Buildable envelope depth: 153 m [fact.envelope-depth]

from fixture: site.depthM

cites: DEMO-B-SETBACK-FRONT, DEMO-B-SETBACK-PERIPHERY

Ground-coverage cap: 12150 sqm [fact.coverage-cap-area]

from fixture: site.widthM, site.depthM

cites: DEMO-B-SITE-COVERAGE-MAX

note: coverage % x gross site area

Density ceiling: 250 DU [fact.density-ceiling]

from fixture: site.widthM, site.depthM

cites: DEMO-B-DENSITY-MAX

note: floor(site acres x density). A ceiling only — NOT a geometric capacity.

Townhouses placed in this reference layout: 50 DU [fact.placed-du]

from fixture: site.widthM, site.depthM

cites: DEMO-B-SETBACK-PERIPHERY, DEMO-B-SETBACK-FRONT, DEMO-B-ROAD-WIDTH-PRIMARY,
DEMO-B-ROAD-WIDTH-SECONDARY, DEMO-B-OPEN-SPACE-MIN, DEMO-B-AMENITY-SHARE-MIN,
DEMO-B-UNIT-PLOT-FRONTAGE-MIN, DEMO-B-UNIT-PLOT-DEPTH-MIN, DEMO-B-ROW-LENGTH-MAX,

DEMO-B-SITE-COVERAGE-MAX, DEMO-B-DENSITY-MAX

note: Counted from canonical geometry. Not a claimed maximum: no completeness search over the layout space was performed.

Shortfall against request: 450 DU [fact.shortfall-du]

from fixture: site.requestedDwellingUnits, site.widthM, site.depthM

cites: DEMO-B-SETBACK-PERIPHERY, DEMO-B-SETBACK-FRONT, DEMO-B-ROAD-WIDTH-PRIMARY, DEMO-B-ROAD-WIDTH-SECONDARY, DEMO-B-OPEN-SPACE-MIN, DEMO-B-AMENITY-SHARE-MIN, DEMO-B-UNIT-PLOT-FRONTAGE-MIN, DEMO-B-UNIT-PLOT-DEPTH-MIN, DEMO-B-ROW-LENGTH-MAX, DEMO-B-SITE-COVERAGE-MAX, DEMO-B-DENSITY-MAX

note: requested - placed

Built footprint (plots + club): 7488 sqm [fact.built-footprint]

from fixture: site.widthM, site.depthM

cites: DEMO-B-UNIT-PLOT-FRONTAGE-MIN, DEMO-B-UNIT-PLOT-DEPTH-MIN, DEMO-B-AMENITY-SHARE-MIN, DEMO-B-SETBACK-PERIPHERY, DEMO-B-SETBACK-FRONT, DEMO-B-ROAD-WIDTH-PRIMARY, DEMO-B-ROAD-WIDTH-SECONDARY, DEMO-B-OPEN-SPACE-MIN, DEMO-B-AMENITY-SHARE-MIN, DEMO-B-UNIT-PLOT-FRONTAGE-MIN, DEMO-B-UNIT-PLOT-DEPTH-MIN, DEMO-B-ROW-LENGTH-MAX, DEMO-B-SITE-COVERAGE-MAX, DEMO-B-DENSITY-MAX

note: Conservative demo convention: each townhouse plot counted fully as footprint.

Green/open space required: 10125 sqm [fact.green-required]

from fixture: site.widthM, site.depthM

cites: DEMO-B-OPEN-SPACE-MIN

Green/open space provided (measured): 10137 sqm [fact.green-provided]

from fixture: site.widthM, site.depthM

cites: DEMO-B-OPEN-SPACE-MIN, DEMO-B-SETBACK-PERIPHERY, DEMO-B-SETBACK-FRONT, DEMO-B-ROAD-WIDTH-PRIMARY

Amenity parcel required: 3240 sqm [fact.amenity-required]

from fixture: site.widthM, site.depthM

cites: DEMO-B-AMENITY-SHARE-MIN

Amenity parcel provided (measured): 3240 sqm [fact.amenity-provided]

from fixture: site.widthM, site.depthM

cites: DEMO-B-AMENITY-SHARE-MIN, DEMO-B-UNIT-PLOT-DEPTH-MIN

Required parking: 75 ECS [fact.parking-required]

from fixture: site.widthM, site.depthM

cites: DEMO-B-PARKING-ECS-PER-DU, DEMO-B-SETBACK-PERIPHERY, DEMO-B-SETBACK-FRONT, DEMO-B-ROAD-WIDTH-PRIMARY, DEMO-B-ROAD-WIDTH-SECONDARY, DEMO-B-OPEN-SPACE-MIN, DEMO-B-AMENITY-SHARE-MIN, DEMO-B-UNIT-PLOT-FRONTAGE-MIN, DEMO-B-UNIT-PLOT-DEPTH-MIN, DEMO-B-ROW-LENGTH-MAX, DEMO-B-SITE-COVERAGE-MAX, DEMO-B-DENSITY-MAX

note: placed DU x parking norm, rounded up. Parking is on-plot (stilt) in v0 and not drawn.

VERDICT — REQUEST vs CEILING vs PLACED

requested [fact.requested-du] ceiling [fact.density-ceiling]

placed [fact.placed-du] shortfall [fact.shortfall-du]

binding: Geometric fill of this reference layout under the cited setback, road, open-space, amenity, plot-dimension and row-length entries. (DEMO-B-SETBACK-PERIPHERY, DEMO-B-SETBACK-FRONT, DEMO-B-ROAD-WIDTH-PRIMARY, DEMO-B-ROAD-WIDTH-SECONDARY, DEMO-B-OPEN-SPACE-MIN, DEMO-B-AMENITY-SHARE-MIN, DEMO-B-UNIT-PLOT-FRONTAGE-MIN, DEMO-B-UNIT-PLOT-DEPTH-MIN, DEMO-B-ROW-LENGTH-MAX)

The client asked for 500 townhouses. Under the DEMO-SLICE-B entries, the density ceiling is 250

DU and 50 townhouses are placed in this reference layout (shortfall 450). The placed count is what this layout achieves; it is not a claimed legal capacity, because no completeness search over the permitted layout space was performed. All values are illustrative demo entries — this is not a statement about any real jurisdiction.

CITATION SNAPSHOT (SELF-CONTAINED — VALID WITHOUT THE RULEBOOK FILES)

DEMO-B-SITE-COVERAGE-MAX slot=site-coverage-max value=30 percent version=DEMO-B-SITE-COVERAGE-MAX#v1
demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Built footprint may cover at most 30% of gross site area (denominator: gross site area).

DEMO-B-DENSITY-MAX slot=density-max value=25 du-per-acre version=DEMO-B-DENSITY-MAX#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: At most 25 dwelling units per

acre of gross site area.

DEMO-B-HEIGHT-MAX slot=height-max value=9 metre version=DEMO-B-HEIGHT-MAX#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: No structure exceeds 9 m above finished ground level.

DEMO-B-STOREYS-MAX slot=storeys-max value=2 storeys version=DEMO-B-STOREYS-MAX#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Townhouses of at most two storeys (G+1).

DEMO-B-SETBACK-PERIPHERY slot=setback-periphery value=12 metre version=DEMO-B-SETBACK-PERIPHERY#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: A 12 m setback from every site boundary that does not face the access road.

DEMO-B-SETBACK-FRONT slot=setback-front value=15 metre version=DEMO-B-SETBACK-FRONT#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: A 15 m setback from the boundary facing the external access road.

DEMO-B-ROAD-WIDTH-PRIMARY slot=road-width-primary value=15 metre version=DEMO-B-ROAD-WIDTH-PRIMARY#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: The internal spine road is 15 m wide, right-of-way.

DEMO-B-ROAD-WIDTH-SECONDARY slot=road-width-secondary value=12 metre version=DEMO-B-ROAD-WIDTH-SECONDARY#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Internal secondary/row roads are 12 m wide, right-of-way.

DEMO-B-OPEN-SPACE-MIN slot=open-space-min value=25 percent version=DEMO-B-OPEN-SPACE-MIN#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: At least 25% of gross site area is green/open space (denominator: gross site area).

DEMO-B-PARKING-ECS-PER-DU slot=parking-ecs-per-du value=1.5 ecs-per-du version=DEMO-B-PARKING-ECS-PER-DU#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: One and a half equivalent car spaces per dwelling unit.

DEMO-B-AMENITY-SHARE-MIN slot=amenity-share-min value=8 percent version=DEMO-B-AMENITY-SHARE-MIN#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: At least 8% of gross site area is the community amenity parcel, club + pool (denominator: gross site area).

DEMO-B-UNIT-PLOT-FRONTAGE-MIN slot=unit-plot-frontage-min value=8 metre version=DEMO-B-UNIT-PLOT-FRONTAGE-MIN#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Each townhouse plot has at least 8 m of road frontage.

DEMO-B-UNIT-PLOT-DEPTH-MIN slot=unit-plot-depth-min value=18 metre version=DEMO-B-UNIT-PLOT-DEPTH-MIN#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Each townhouse plot is at least 18 m deep.

DEMO-B-ROW-LENGTH-MAX slot=row-length-max value=48 metre version=DEMO-B-ROW-LENGTH-MAX#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: One unbroken townhouse row is at most 48 m long.

NOTES

DEMO: every value is illustrative and unverified; no real authority, document, or jurisdiction is represented.

Percent rules use gross site area as denominator, stated per entry; provided areas are measured from canonical geometry.

The requested unit count is client intent and is never copied into any result field.

Height and storey caps are cited envelope limits; v0 draws no elevations, and no drawing label exceeds them.